VECTOR SPACES · DIAGNOSTICS

Four questions, one reduction

Each question sets up a matrix, reduces it, and reads a different part of the result. The setup barely changes between them — what changes is where to look.

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QUESTIONS

ABOUT A SINGLE VECTOR 1

1 **Is b in the span?**

`consistent ⇒ yes`

$[\,\mathbf{v}_1 \cdots \mathbf{v}_k \mid \mathbf{b}\,]$

Augment the spanning vectors with $\mathbf{b}$ and reduce. A row of the form $[\,0 \cdots 0 \mid d\,]$ with $d \neq 0$ is a contradiction and settles it as no; anything else means $\mathbf{b}$ is reachable, and the reduced form also hands you the coefficients that reach it.

ABOUT THE WHOLE SET 3

2 **Does the set span $\mathbb{R}^n$?**

`a pivot in every row`

$\operatorname{rank}(A) = n$

Vectors as columns of an $n \times k$ matrix. Every row needs a pivot — a pivotless row is a direction of $\mathbb{R}^n$ nothing in the set reaches. Note this is a condition on rows, while redundancy below is a condition on columns; confusing the two is the usual error.

3 **Which vectors are redundant?**

`a pivot in every column ⇒ none`

columns of A at the non-pivot positions of $\operatorname{rref}(A)$ (one valid choice)

A column without a pivot corresponds to a vector already in the span of the earlier ones, so it can be dropped without changing the span. The reduced form names which — removing them leaves a minimal spanning set.

4 **Is the set a basis?**

`a pivot in every row and column`

$\det(A) \neq 0$, A square

Both conditions at once: spanning with no redundancy. For n vectors in $\mathbb{R}^n$ the matrix is square, so the determinant test settles it in one step — which is the invertibility equivalence restated for spanning sets.

The common step is worth stating plainly: build a matrix from the vectors, reduce, inspect. Membership reads the augmented column, spanning reads the rows, redundancy reads the columns, and a basis needs both. Nothing here requires a technique the reduction does not already provide.